

Intrusion Prevention Systems (IPSs)

Learning objectives

- **Explain the role and function of an Intrusion Prevention System (IPS) in network security**
- **Compare IPS with IDS and describe the key differences between them**
- **Identify how IPS detects and prevents threats using various detection methods**

What is an IPS?

- **IPS = Intrusion Prevention System**
- **Monitors network traffic in real time**
- **Detects malicious activity**
- **Automatically blocks, drops, or resets connections**
- **Works inline with network flow**

IPS vs IDS: what's the difference?

IPS (Intrusion Prevention System)

- Monitors network traffic
- **Actively blocks** malicious traffic
- Prevents threats in real time
- Works **inline** (directly in traffic flow)
- **Higher risk** of false positives if not properly configured
- Analogy: Like a **security guard** who stops intruders

IDS (Intrusion Detection System)

- Monitors network traffic
- Detects threats and **raises alerts only**
- **Does not block** or stop attacks
- Works **out of line** (passive monitoring)
- **Low risk** of false positives
- Analogy: Like a **security camera** that watches but doesn't act

How does an IPS work?

- **Analyses packets in real time**
- **Compares them to known signatures or behaviour patterns**
- **Takes action based on rules**
- **Logs incidents and alerts administrators**

How does IPS detect threats?

- **Signature-based: matches known patterns (e.g. malware)**
- **Anomaly-based: flags unusual behaviour (e.g. sudden traffic spikes)**
- **Policy-based: uses administrator-defined rules**
- **Hybrid: combines multiple methods for better accuracy**

Where is IPS placed in a network?

- **Inline between firewall and internal network**
- **At key entry points (e.g. data centre, VPN, cloud gateway)**
- **Integrated into Next-Generation Firewalls (NGFW)**

Why use IPS?

- **Stops threats before they spread**
- **Blocks known malware and exploits**
- **Reduces load on internal systems**
- **Helps meet compliance (e.g. GDPR, PCI DSS)**
- **Boosts overall network visibility**

The limitations of IPS

- **False positives may block legitimate traffic**
- **Requires ongoing updates and tuning**
- **Can be a bottleneck if not properly sized**
- **Doesn't replace all other defences**

A quick recap

- **IPS = real-time prevention system**
- **Works inline to stop attacks**
- **Uses signatures, anomalies, and rules**
- **Needs maintenance and careful configuration**
- **Part of a layered defence strategy**

Reflection

- **Would you rather deploy IDS, IPS or both in a new network?**
- **Why?**

Next: access control models

- **Mandatory access control models**
- **How access control supports cybersecurity**
- **Learn how systems decide who gets access to what, and why it matters**